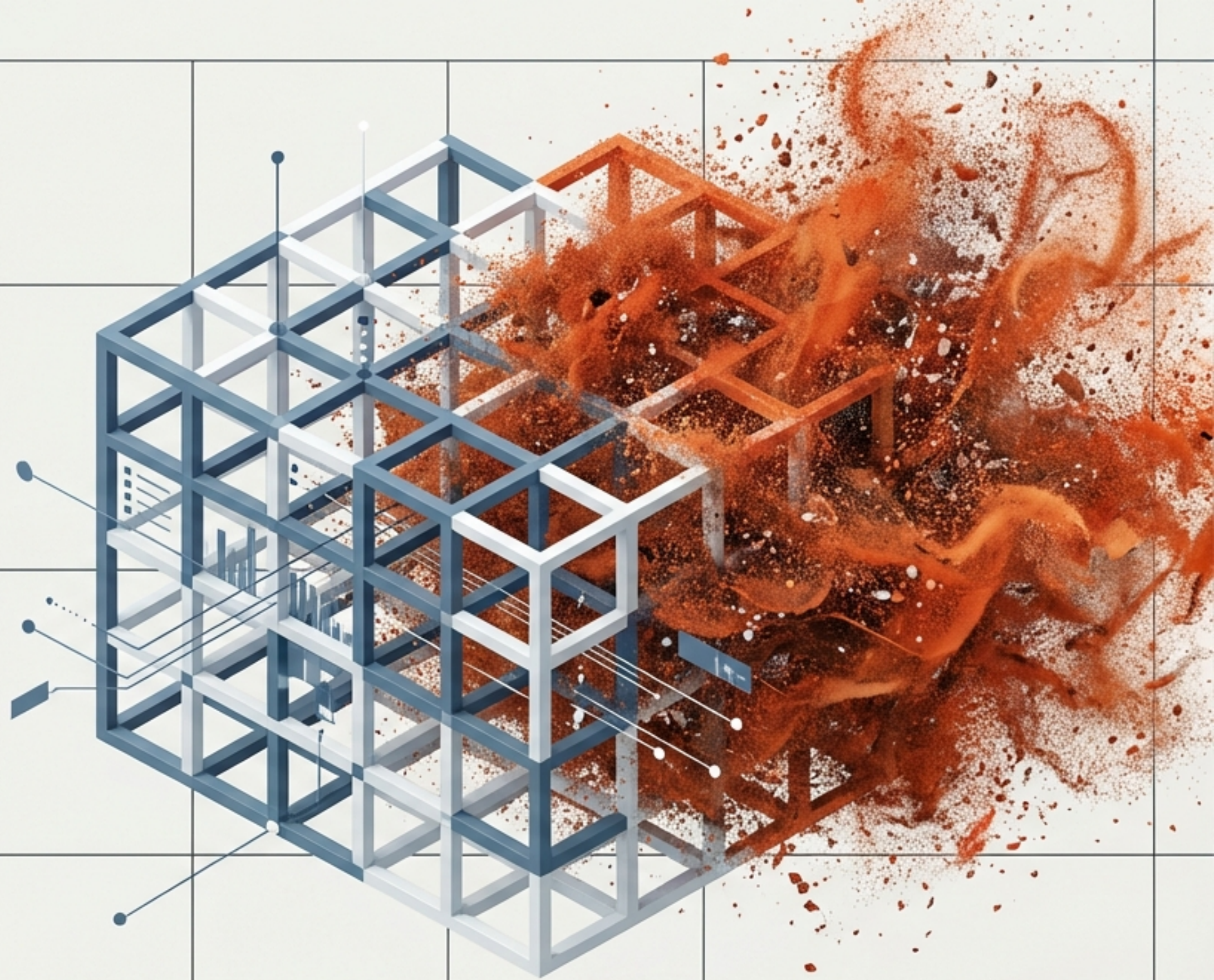


Signals in the Noise

How modern engineering measures, filters, and harnesses entropy across AI, cybersecurity, and physical systems.



The Entropy Paradox

Historically, noise was a hurdle to be eliminated. Today, it is a diagnostic tool, a computational resource, and a mathematical filter.

Measuring & Hunting

Information Theory &
IoT Sensor Networks

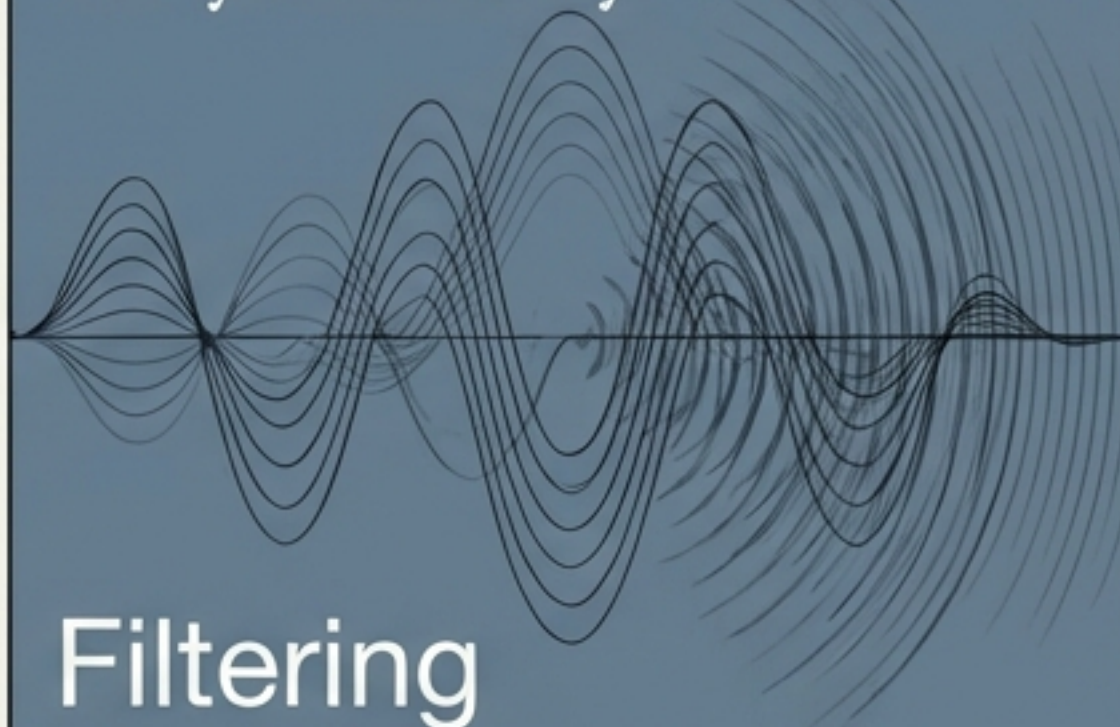


Measuring & Hunting

Information Theory &
IoT Sensor Networks

Filtering

Acoustic Signal Processing
& Cybersecurity

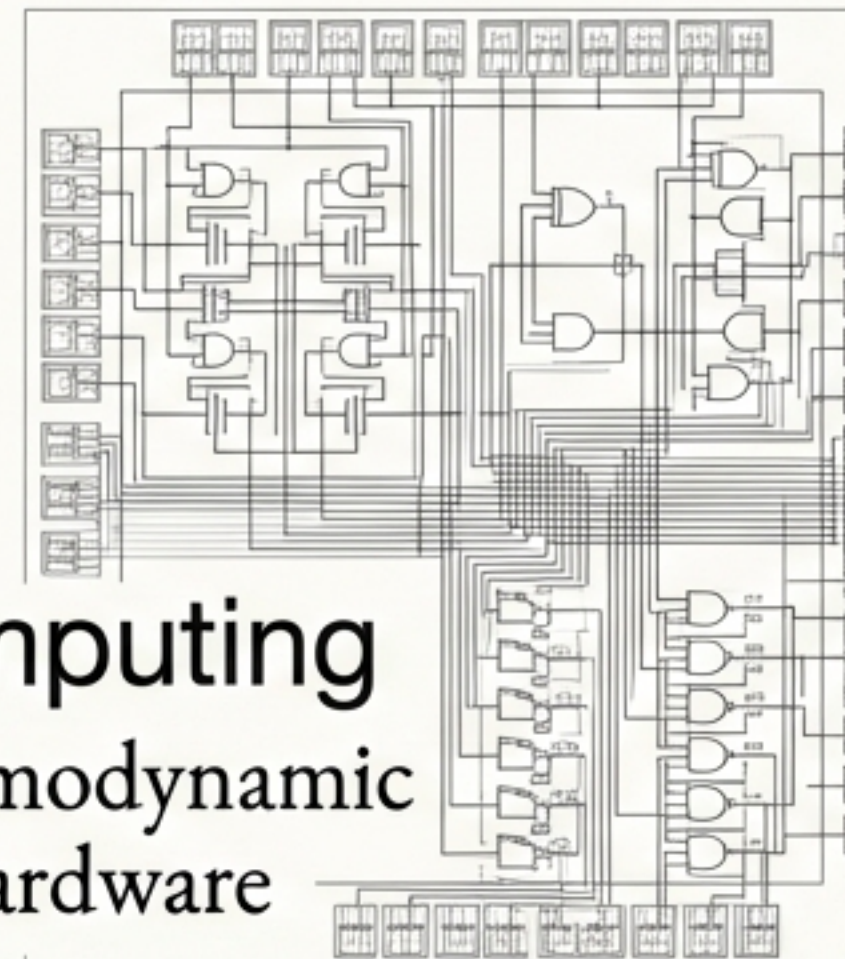


Filtering

Acoustic Signal Processing
& Cybersecurity

Computing

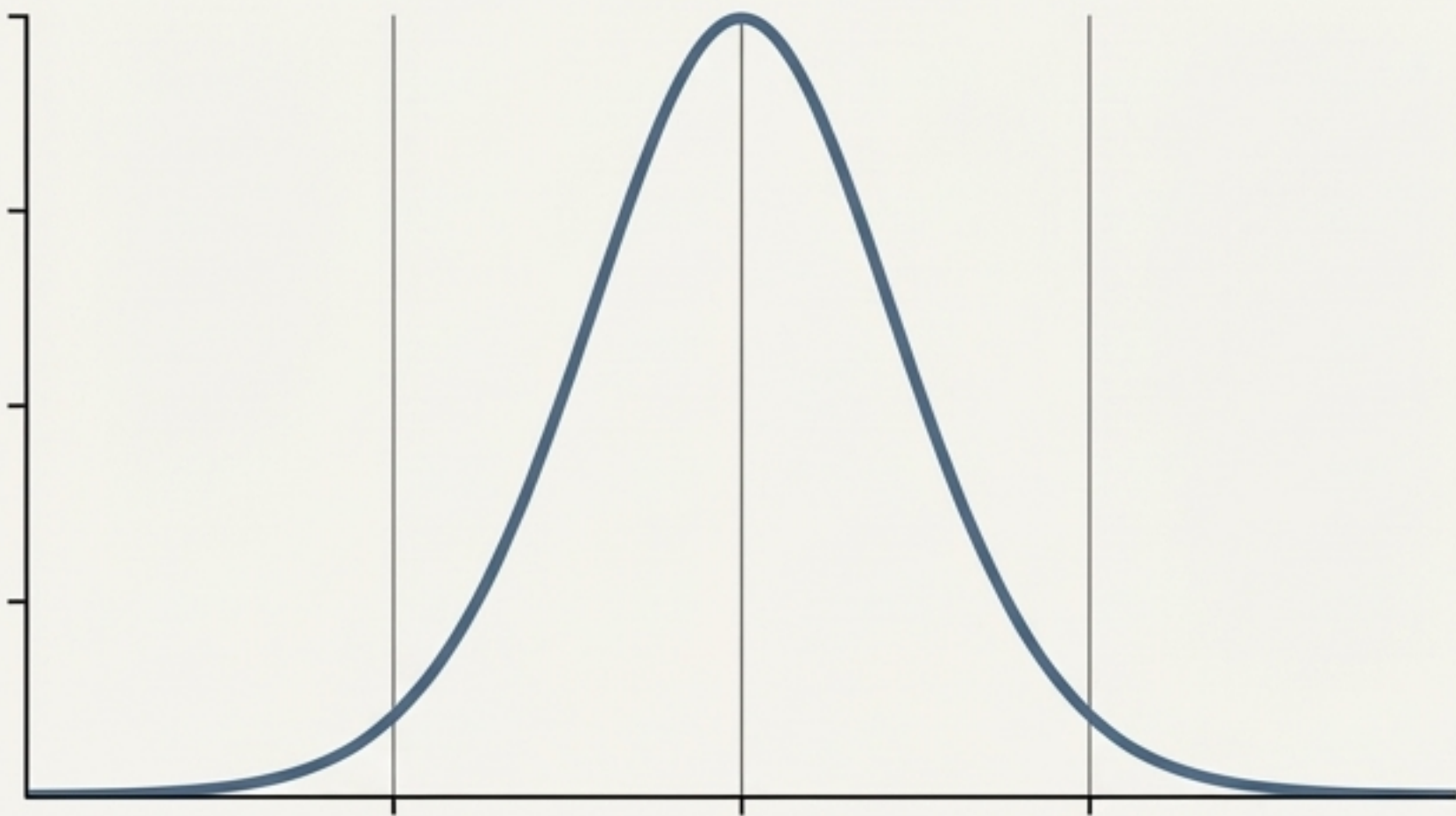
Thermodynamic
AI Hardware



Computing

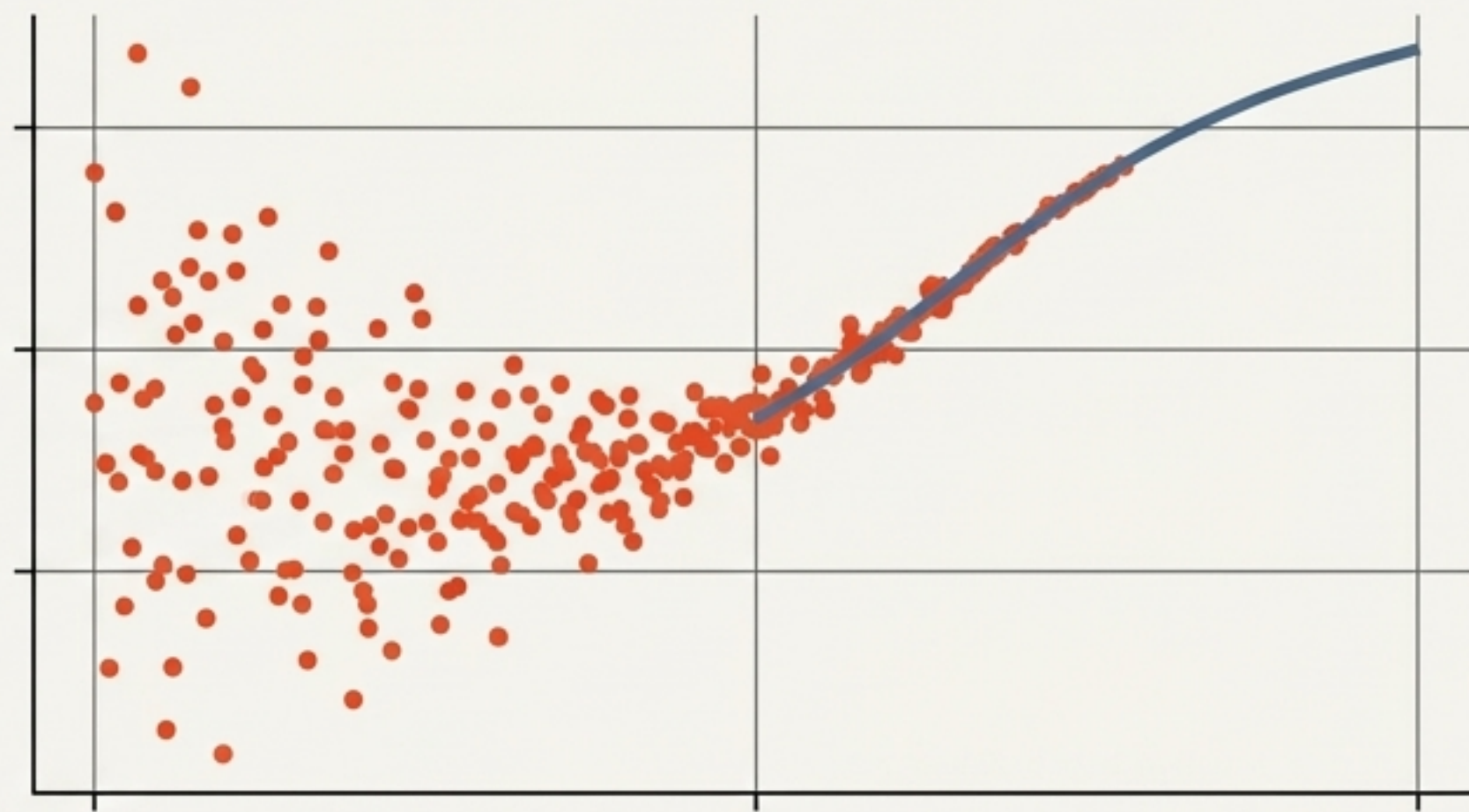
Thermodynamic
AI Hardware

The Mathematics of the Unknown



Parametric Estimation

Assumes a strict mathematical model (Gaussian, Markov, Renewal). Calculates entropy rate via plug-in estimation.



Non-Parametric Estimation

Data-driven, relying on limit theorems (Lempel-Ziv, Approximate Entropy).

Key Takeaway: Fewer assumptions lead to greater robustness. Non-parametric methods allow complex stochastic processes to reveal their own underlying uncertainty, $H(\chi)$.

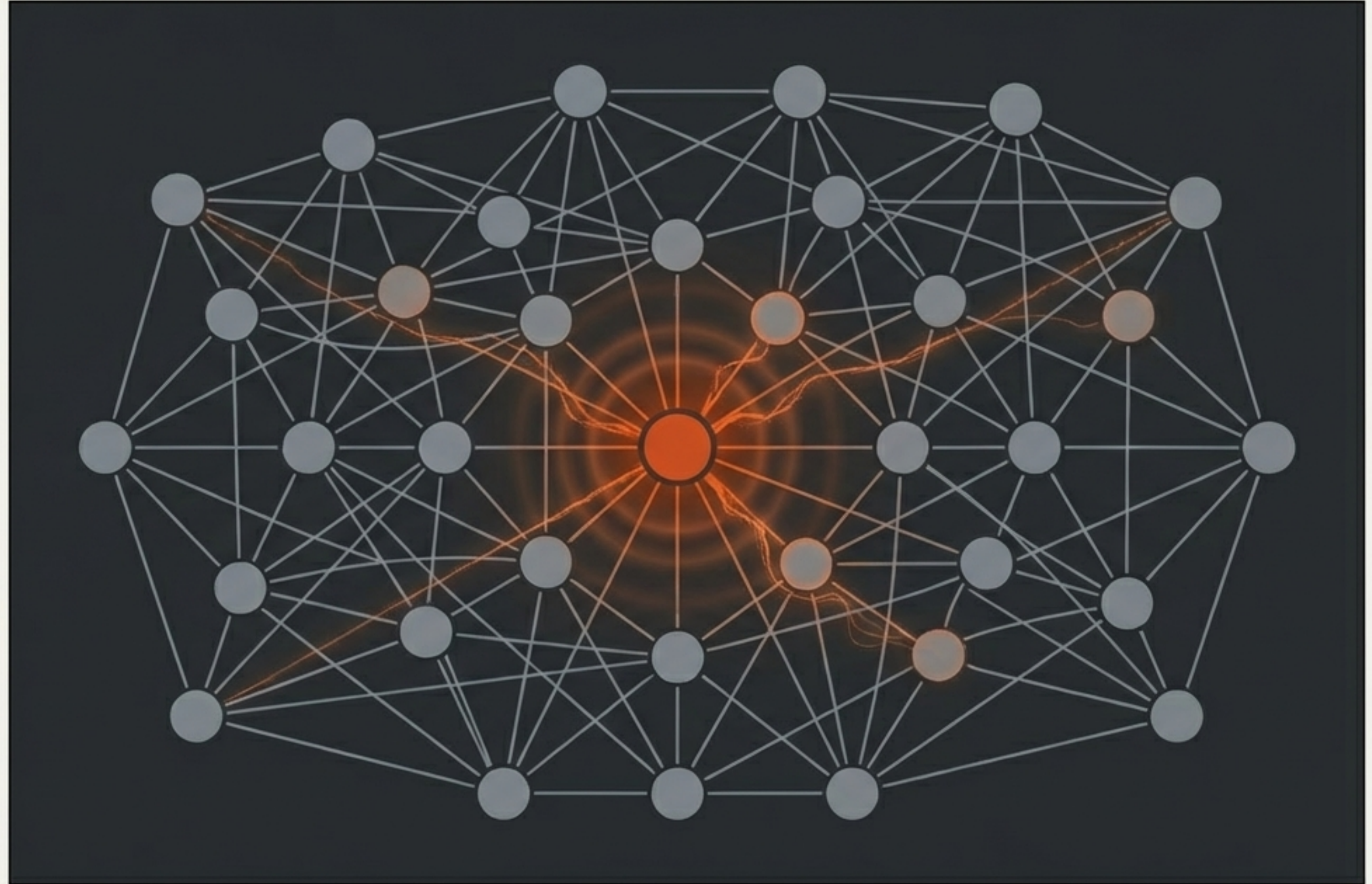
The Complex Systems Challenge

Context

In Cyber-Physical Systems (e.g., Mars Rovers, Water Treatment facilities), local anomalies cascade through hidden structural connections.

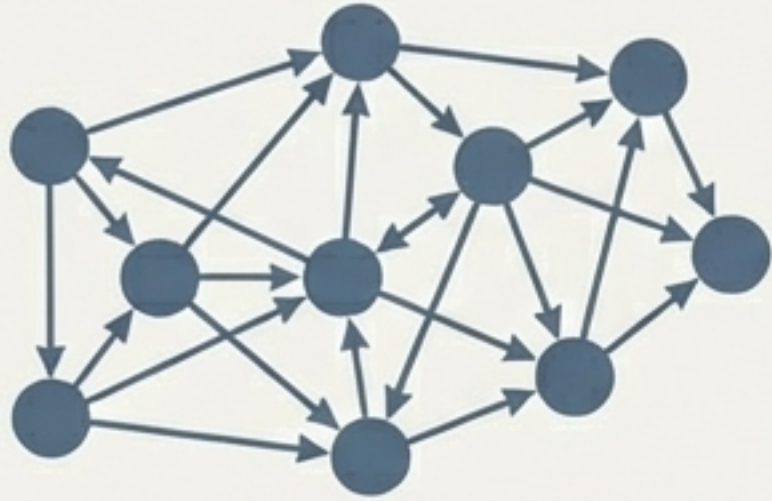
The Flaw

Standard multivariate time-series analysis fails when this structural coupling is ignored. You cannot treat sensors in isolation.



Entropy-Based Feature Aggregation

Step 1: Sensor Graph Generation



Step 1: Sensor Graph Generation

Maps multi-element relationships and structural dependencies.

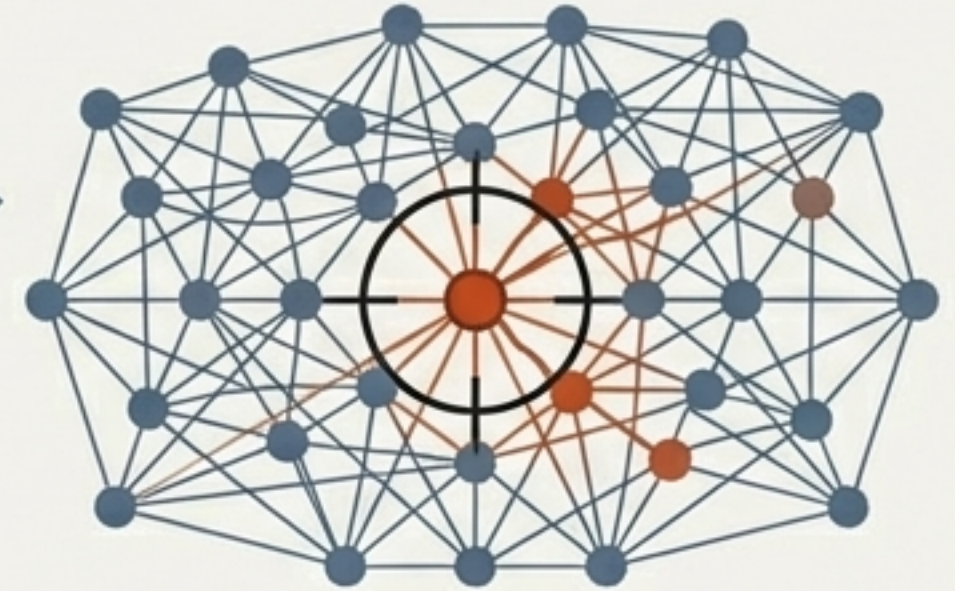
Step 2: Entropy Centrality Attention

$$H(v_i) = H(G) - H(G \setminus \{v_i\})$$

Step 2: Entropy Centrality Attention

Removing a critical node decreases system entropy the most.

Step 3: Anomaly Detection

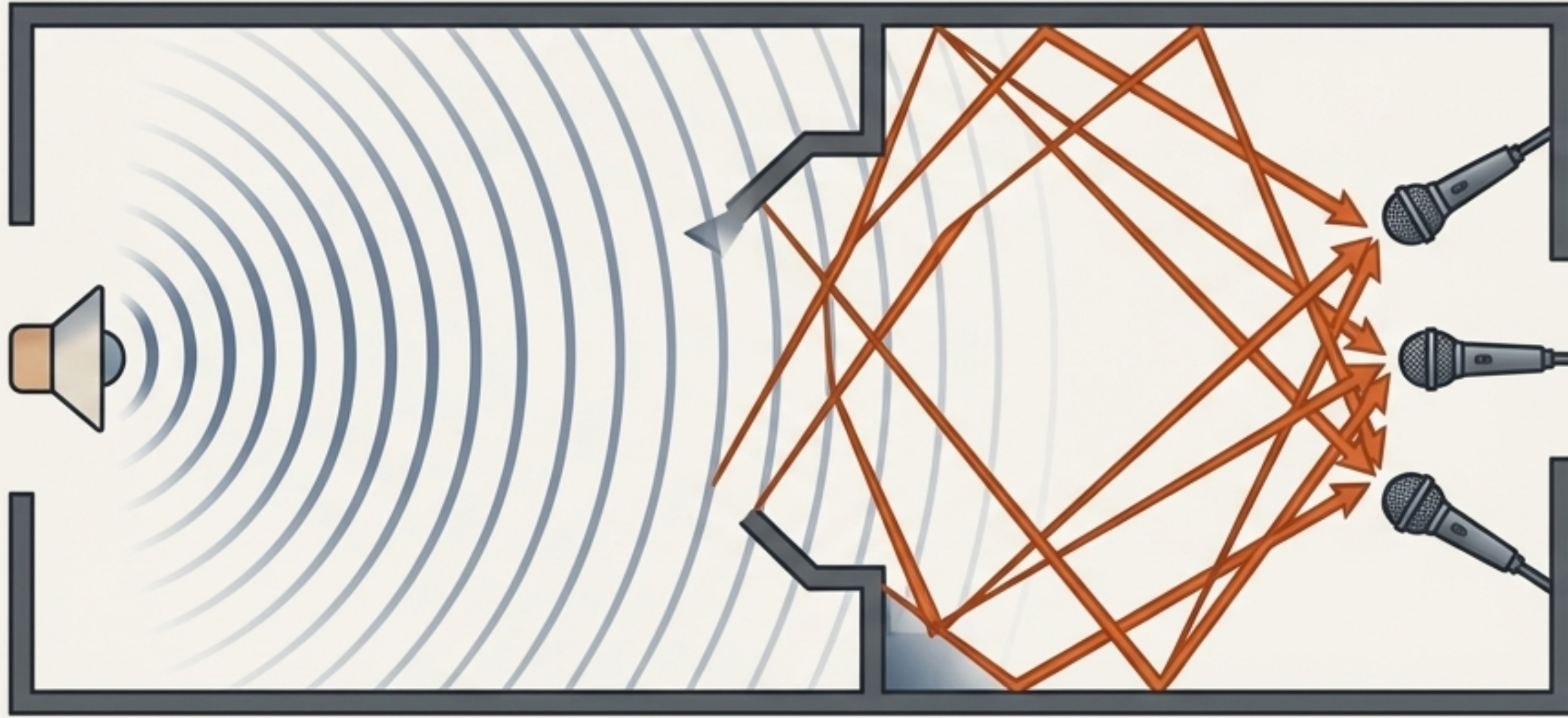


Step 3: Anomaly

Isolated fault or anomaly indicating that the entropy detection.

Applying this entropy-aware GNN approach yields massive F1-Score performance gains on real-world datasets like MSL (Mars Science Laboratory) and WADI.

The Time Delay Estimation Problem



The Goal

Locate a radiating source in a highly reverberant, noisy room.

The Hurdle

The industry-standard Generalized Cross-Correlation (GCC) method breaks down when reverberation is high and the noise is non-Gaussian.

Minimum Joint Entropy vs. Cross-Correlation

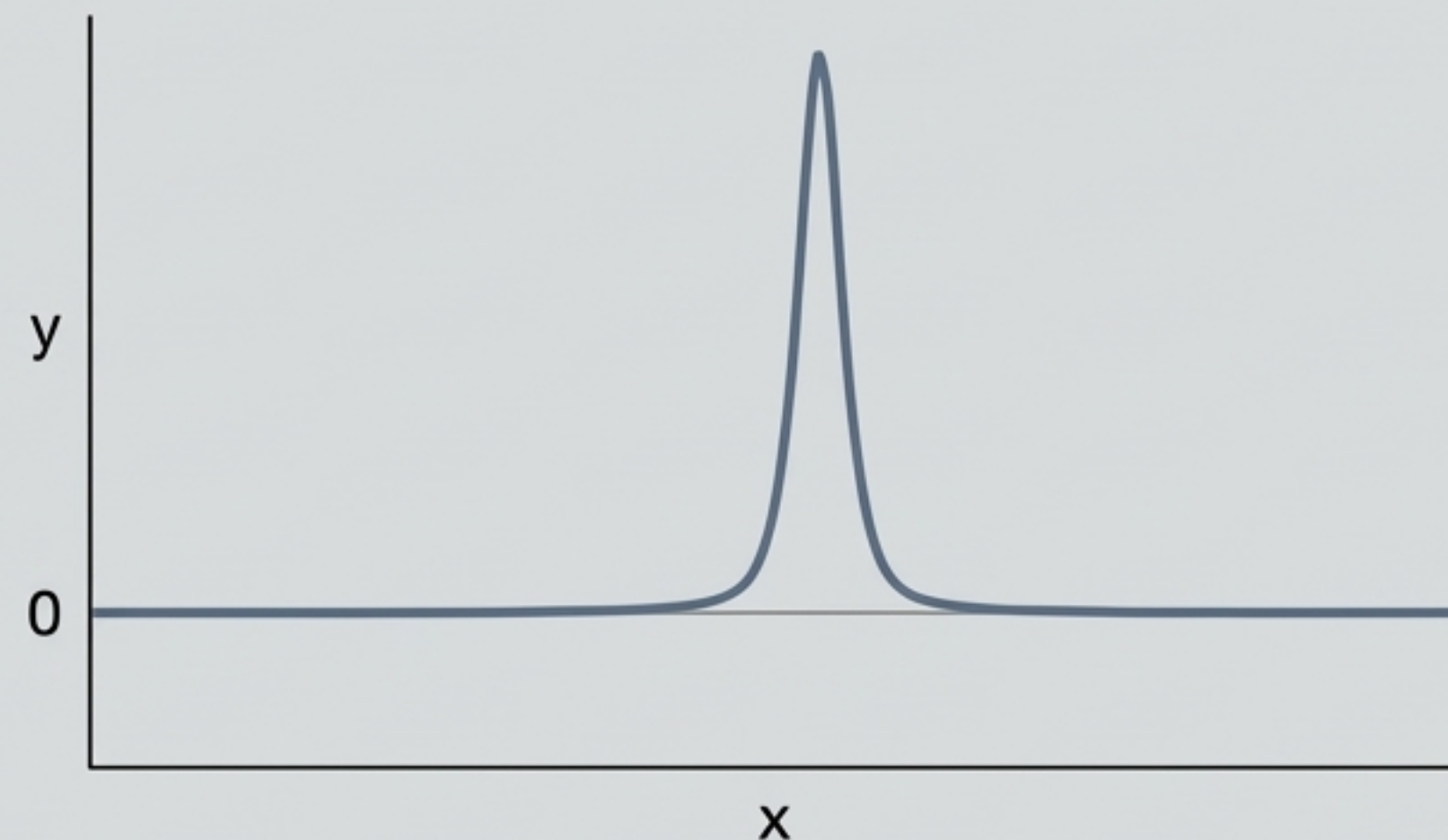
The Failure: Multichannel Cross-Correlation Coefficient (MCCC)

Fails in high reverberation (Bell Labs Varechoic chamber).



The Solution: Minimum Entropy TDE

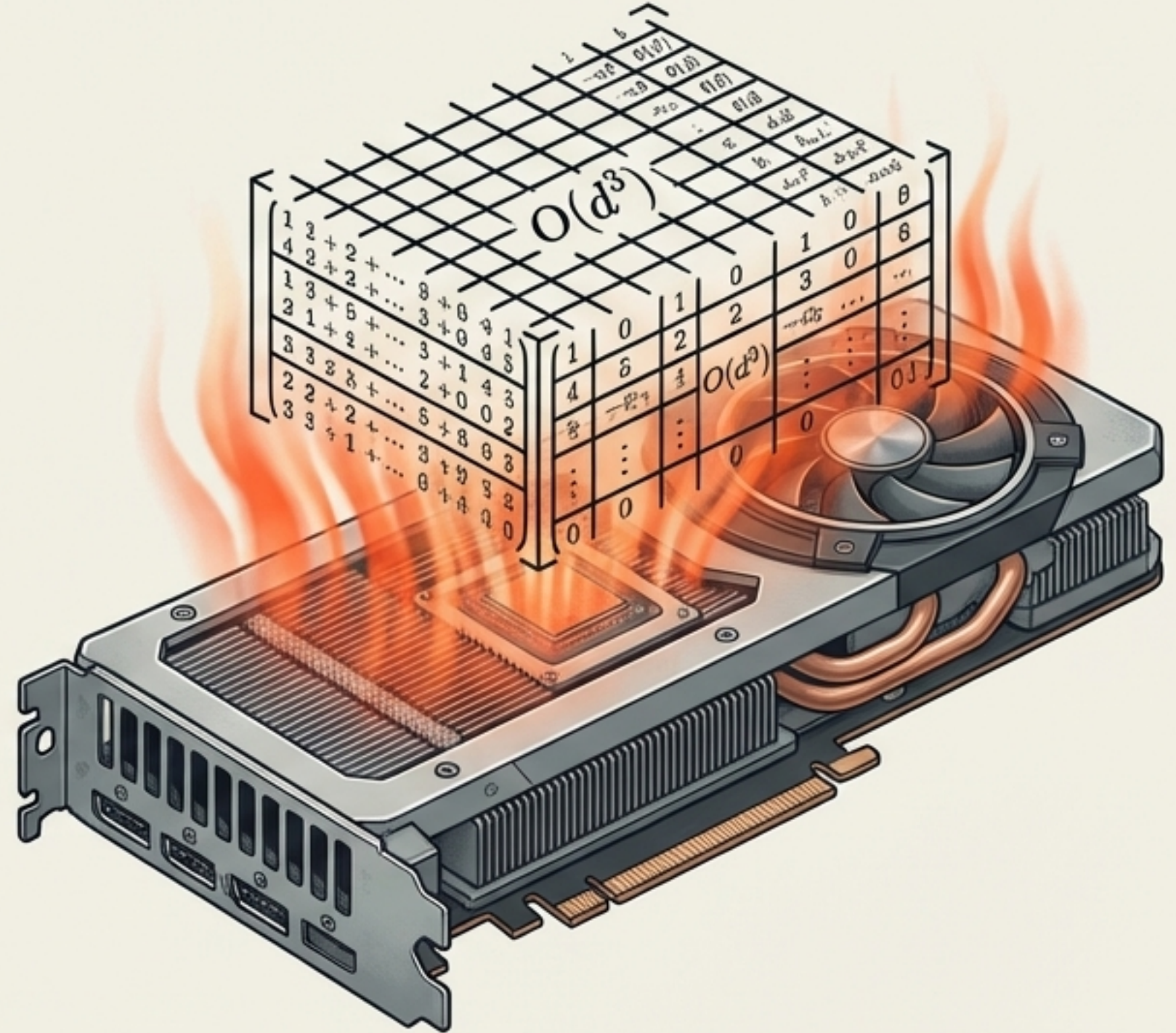
Models speech correctly to sharpen the signal estimate.



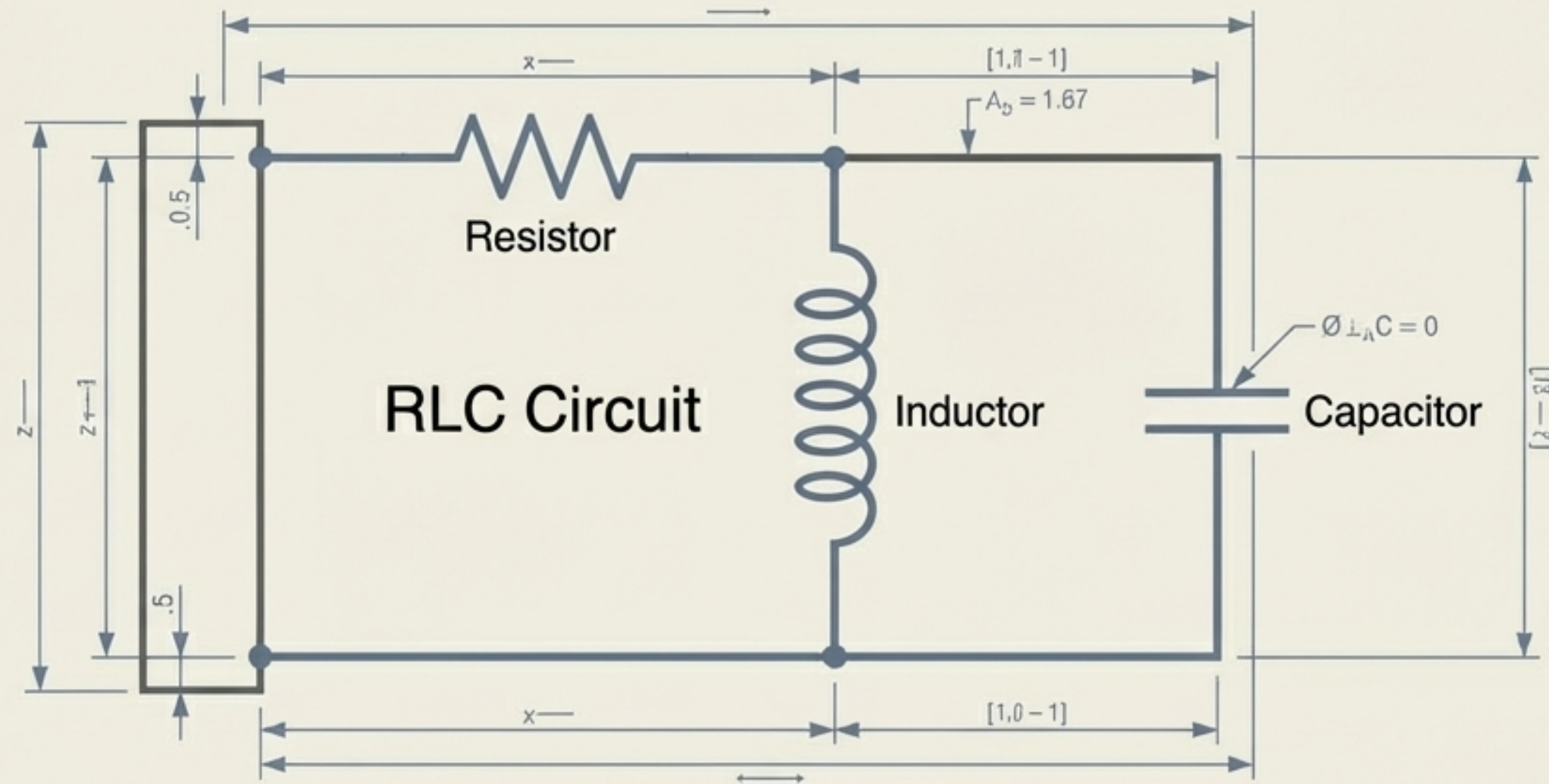
MCCC assumes Gaussian signals. Human speech is not Gaussian. By modelling speech as a Laplace distribution ($p(x) = \frac{1}{2b} \exp\left(-\frac{|x-\mu|}{b}\right)$), minimising joint entropy successfully filters the multipath chaos where correlation fails.

The Digital AI Bottleneck

- Probabilistic AI (diffusion models, Bayesian inference) demands massive Gaussian sampling and matrix inversion.
- **Standard** digital hardware is fundamentally mismatched for stochastic mathematics.
- **The Cost:** Extreme energy consumption and severe computational bottlenecks scaling at $O(d^3)$.



The Stochastic Processing Unit (SPU)

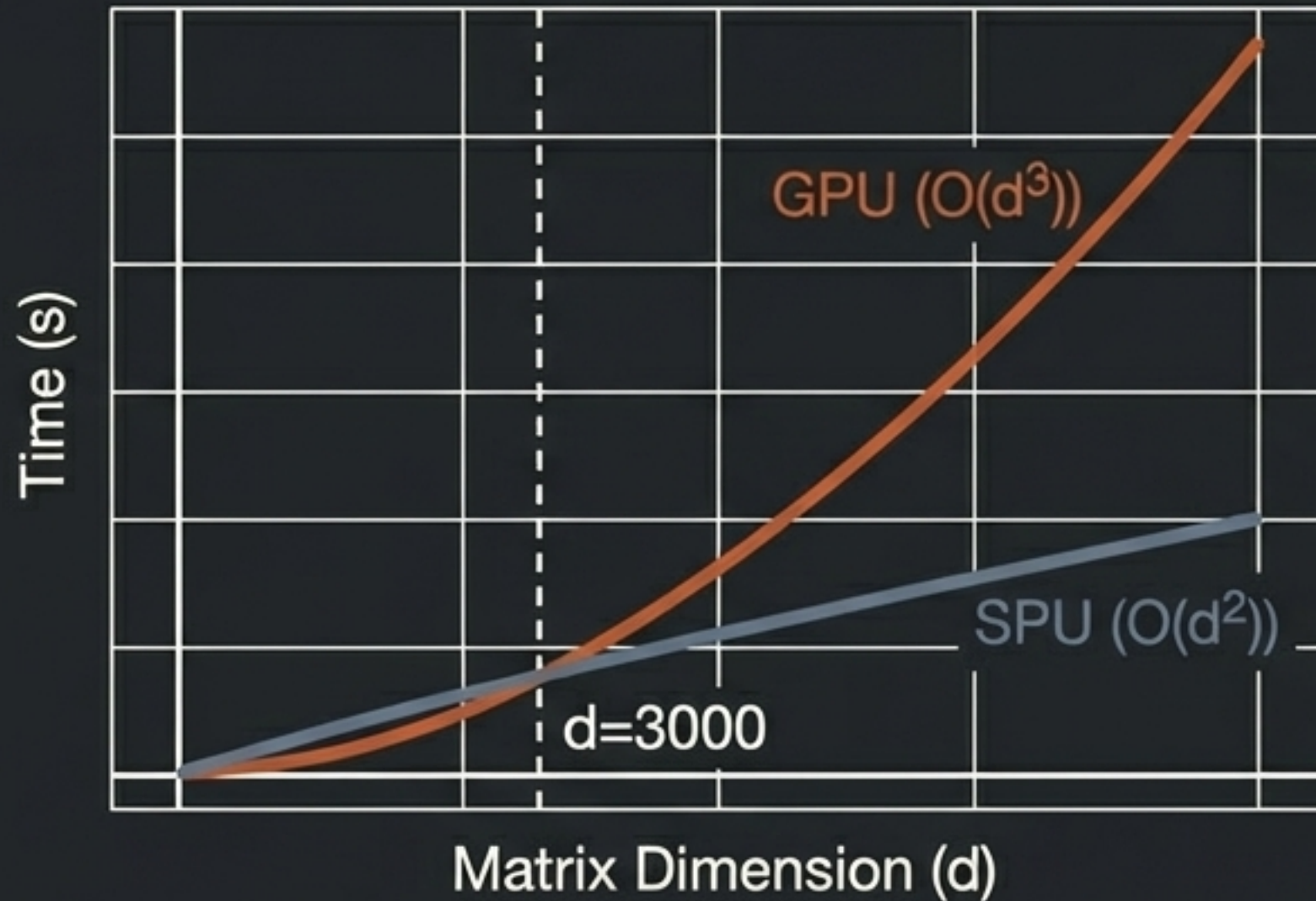


Instead of fighting noise, thermodynamic computing uses physical thermal noise (Langevin dynamics) to compute.

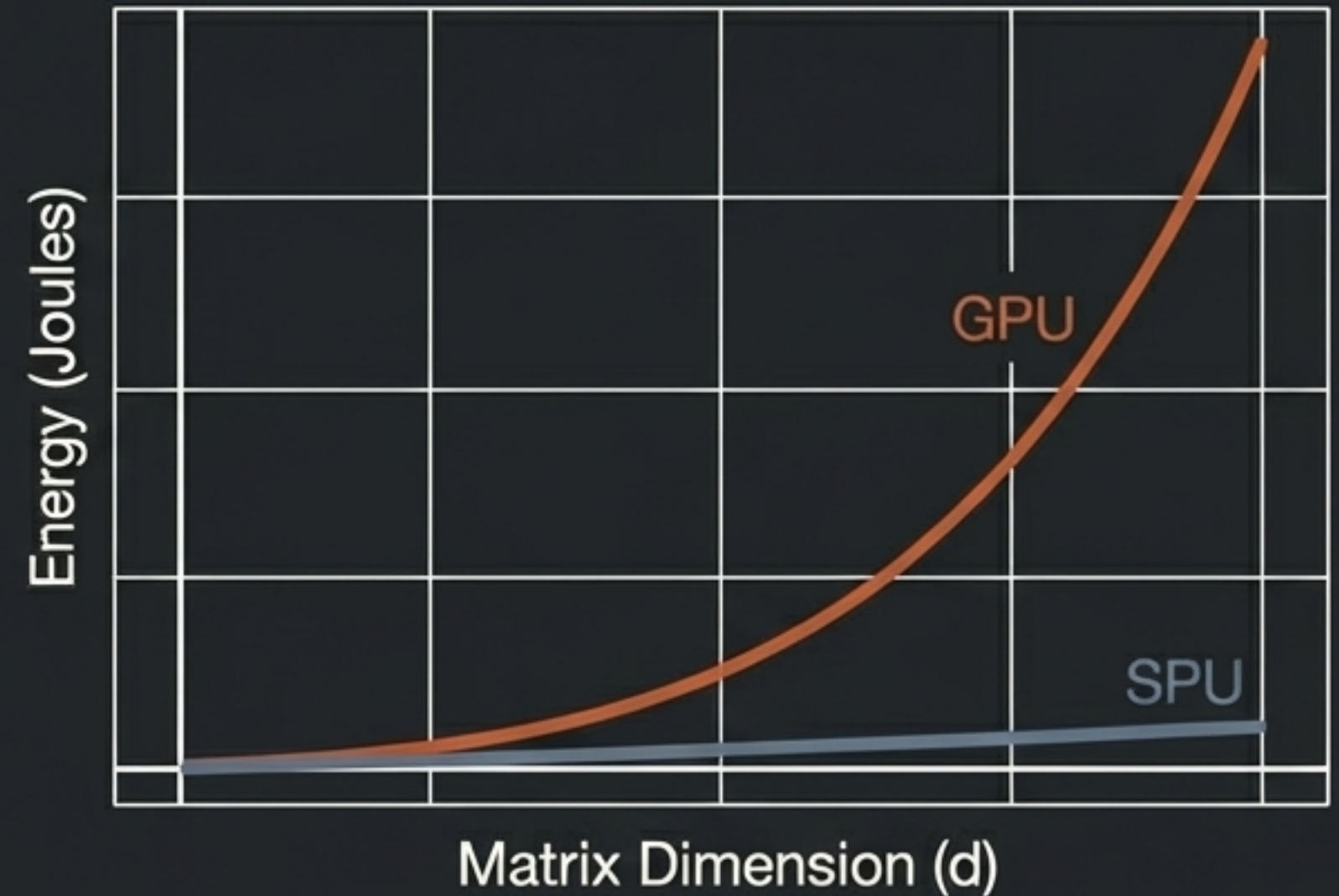
By programming the Maxwell capacitance matrix, the circuit's natural equilibrium state perfectly outputs Gaussian samples and inverted matrices ($\mathbf{v} \sim \mathcal{N}(0, A^{-1})$).

Thermodynamic Advantage at Scale

Time-to-Solution



Energy Consumption



At dimension $d=10,000$, the SPU predicts an order-of-magnitude speedup and massive energy savings over a state-of-the-art NVIDIA A6000 GPU.



The Digital Firehose

Capturing packets is trivially easy. The air is already screaming with data. Reading them is an art.

The Sanity Check

- **File size:** A 4-hour capture at 24 KB means you failed.
- **Format:** PCAP vs raw binary.
- **Encryption reality:** WPA3 without a captured handshake is just encrypted static. A box of locks without keys.

Filtering for Truth

```
wlan.fc.type_subtype == 0x08
```

Beacons. Access points announcing their existence before anyone connects.

```
wlan.fc.type_subtype == 0x04
```

Probes. Phones shouting old network names into the void. Where privacy dies quietly.

```
eapol
```

The Handshake. Four packets in sequence. Without this, WPA cracking is mere theatre.

Reconstructing the Narrative

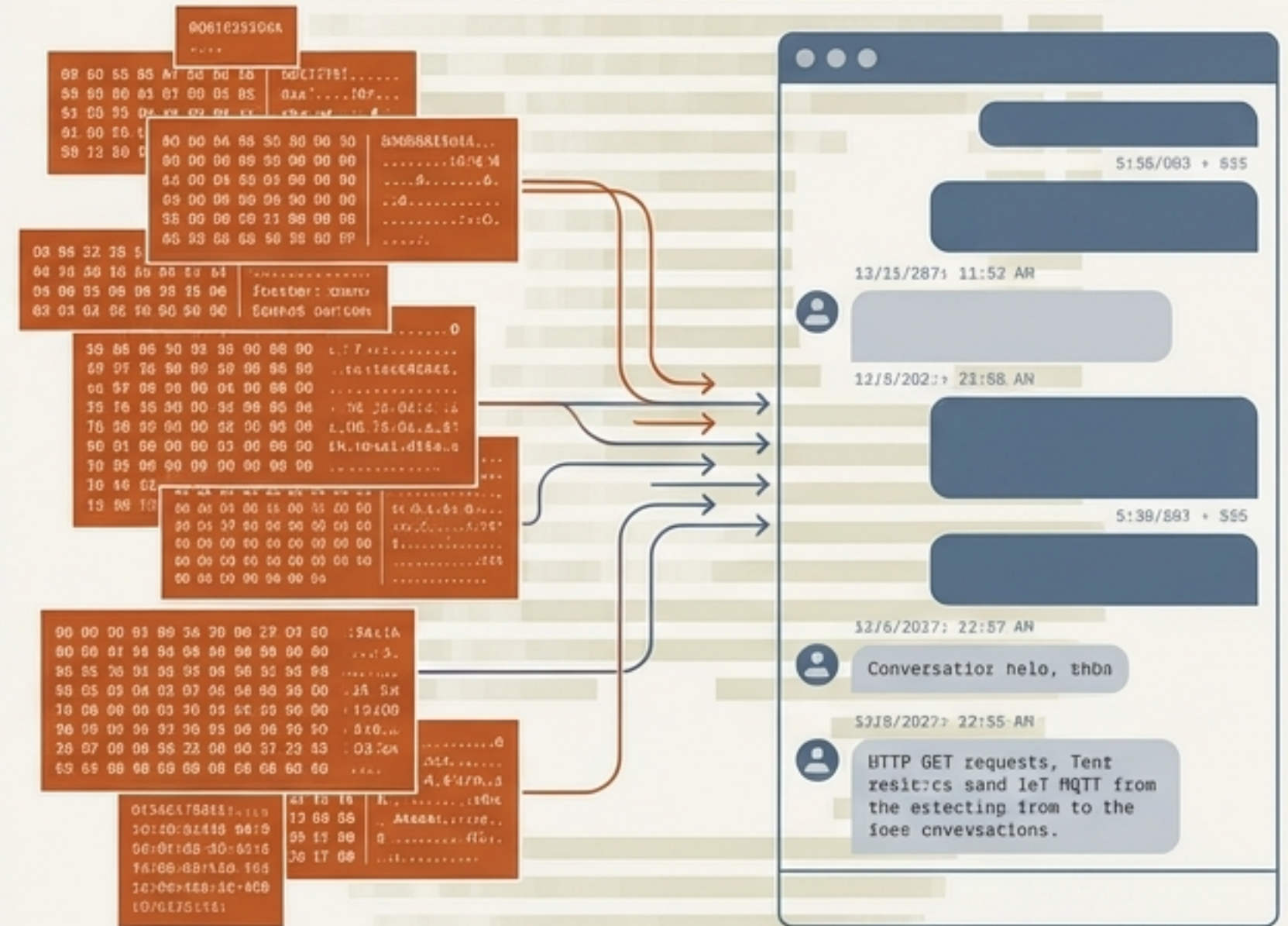
Packets lie when isolated;
streams tell the truth.

Follow the TCP Stream

Order is restored. Malware C2 traffic and
exfiltration patterns emerge.

Cleartext Ghosts

HTTP GET requests, Telnet relics, and IoT MQTT
chatter. Extracting the story of human behaviour
directly from the hex noise.



The Unifying Theory of Noise

Information Theory

Mathematically quantifying the limits of uncertainty.

Cyber-Physical Systems

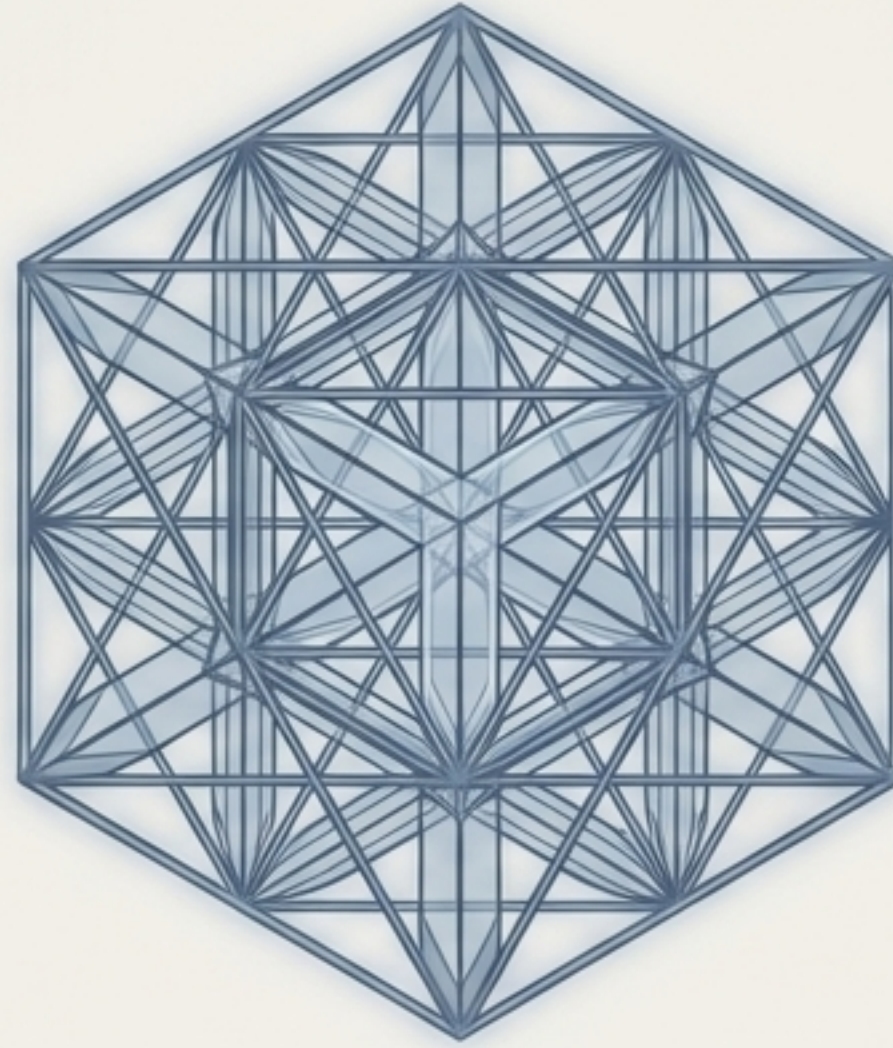
Hunting entropy drops to find mechanical failure.

Acoustics & Cyber

Filtering out the chaotic multipath/hex noise to reveal human intent.

Thermodynamic AI

Harnessing thermal chaos directly as a computational engine.



Master the noise. Master the system.

Entropy is no longer just a measure of decay. It is the raw material of modern engineering.